

2.6 Independent events

Definition 2.7. Two events $A, B \subseteq S$ are **independent** if $P(A \cap B) = P(A)P(B)$.

This seems to come out of nowhere, but it has a nice interpretation. Consider two independent events $A, B \subseteq S$. If you observe that B occurs, then the conditional probability of A is

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A). \quad (2.10)$$

So your probability assessment of the event A was not changed by the news that B occurred. These events “live separate lives,” and knowledge of one does not teach you anything about the likelihood of the other. Back in Example 2.13 and Figure 2.13, this was the case when $\text{TPR} = \text{TNR} = 0.5$. Imagine that the diagnostic test is just the doctor flipping a coin without even examining you. So the test result has absolutely nothing to do with whether or not you have the disease, and $P(D = + | T = +) = P(D = +)$. The probability is the same coming out as it was going in. You learned nothing.

Remark 2.13. Independence is a symmetric concept, so if $P(A | B) = P(A)$, then $P(B | A) = P(B)$ by the same reasoning. If knowing A teaches you nothing about B , then knowing B doesn’t teach you anything about A either.

Example 2.14. Recall the example of rolling two fair, six-sided die. The outcomes are visualized in Table 2.2. Next consider the following events:

A = “Die 1 is less than or equal to 3”

B = “Die 2 is equal to 1”

C = “The sum of the rolls is 10”

D = “The sum of the rolls is 7”

E = “The rolls are the same (doubles)”

F = “The sum is even”.

Since the dice are fair, outcomes are equally likely, so by inspecting Table 2.2 and counting, we know that $P(A) = 18/36 = 1/2$, $P(B) = 6/36$, $P(C) = 3/36$, $P(D) = 6/36$, $P(E) = 6/36$, and $P(F) = 18/36 = 1/2$. Furthermore, we see that $P(A \cap B) = 3/36$, $P(A \cap C) = 0$, $P(A \cap D) = 3/36$, and $P(E \cap F) = 6/36$. As a consequence, we see that

$$\begin{aligned} P(A | B) &= \frac{P(A \cap B)}{P(B)} = \frac{3/36}{6/36} = 1/2 = P(A) \\ P(A | C) &= \frac{P(A \cap C)}{P(C)} = \frac{0}{3/36} = 0 \neq P(A) \\ P(A | D) &= \frac{P(A \cap D)}{P(D)} = \frac{3/36}{6/36} = 1/2 = P(A) \\ P(E | F) &= \frac{P(E \cap F)}{P(F)} = \frac{6/36}{18/36} = 1/3 \neq P(E) \\ P(F | E) &= \frac{P(E \cap F)}{P(E)} = \frac{6/36}{6/36} = 1 \neq P(F). \end{aligned}$$

So A and B are independent events. This makes sense because the dice are rolled separately and their outcomes should not affect one another. So knowing that Die 2 is a 1 should not teach you anything about Die 1. A and C are dependent events, because they are mutually exclusive. If you know that you rolls sum to ten, that guarantees that Die 1 is greater than a three. A and D are independent. If you roll a double, that necessarily means that your sum is also even, so in fact $E \subseteq F$, and these events are quite *dependent*.

Example 2.15. Sally Clark (1964 - 2007) was an English solicitor. In 1996, her first son Christopher died in bed at 11 weeks. In 1998, her second son Harry died in bed at 8 weeks. Clark was charged with murdering them. On both occasions, Clark was at home alone with her baby and there was evidence of trauma (which could have been related to attempts to resuscitate them). Her defense argued that the death's were due to sudden infant death syndrome (SIDS). This is a “diagnosis of exclusion.” Having ruled out every other medically plausible cause, you say “whelp, I guess it must be SIDS then.” The prosecution called pediatrician Roy Meadow to testify as an expert witness, and he claimed that there was a “1 in 73,000,000” chance of two SIDS deaths in the same family. Where did he get that number? Based on the literature at the time, the incidence of SIDS in the UK was approximately 1 in 8,500 in middle-class families with no known risk factors. Meadow made the assumption that the two boys’ deaths were independent, and so he reported:

$$\begin{aligned} P(\text{SIDS} \rightarrow \text{Chris} \cap \text{SIDS} \rightarrow \text{Harry}) &= P(\text{SIDS} \rightarrow \text{Chris})P(\text{SIDS} \rightarrow \text{Harry}) \\ &= \frac{1}{8500} \times \frac{1}{8500} \\ &= \frac{1}{72,250,000}. \end{aligned}$$

This number is simply wrong. The two events are *not* independent. These boys share parents, environment, genetics, and a host of other relevant variables that bear on their health. If we learn something about the health of one of the boys (such as...he died of SIDS), it is insane to think that this would not teach us anything about the other boy’s health.

Having said that, even if Dr. Meadow had computed this probability perfectly correctly, it isn’t actually relevant in this case. But more on that later.

Remark 2.14. Something that inevitably confuses folks is the difference between disjoint and independent events. The colloquial meanings of those words make it feel like the concepts would be similar, but they are actually quite different. Recall:

- **Disjoint events:** they cannot occur at the same time, so $A \cap B = \emptyset$;
- **Independent events:** knowledge of one does not teach you anything about the other, so $P(A | B) = P(A)$.

Right off the bat, one difference we see is that disjointedness is a property of the sets themselves. Independence is a property of the probability measure and how it factors. If you’re asked “are these events disjoint,” you just look at them. You don’t need to know anything about the probabilities to answer that question. If asked “are these events independent,” it’s not enough to know the contents of the sets. You need to know what probability measure is being used. Two events that are independent with respect to one probability measure may not be independent under another. But if they’re disjoint, then they’re always disjoint no matter what the probabilities are.

Here is another illustration of the difference. Say you had two disjoint events A and B , and $P(A) > 0$. If you compute the conditional probability of one given the other, you get

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\emptyset)}{P(B)} = \frac{0}{P(B)} = 0 \neq P(A).$$

So if A and B are disjoint, and I know B occurs, then this *guarantees* that A does not, because disjoint means that events cannot happen at the same time. It's either one or the other. Therefore, instead of teaching us nothing about A , knowing B teaches us *everything* about it. In this way we see that disjointedness is in fact a rather severe form of *dependence*.

Remark 2.15. Can we visualize independent events with a Venn diagram? No. And the reason again is that independence is not fundamentally a property of the sets themselves, but a property of the measure P that is acting on the sets. Visualizing P is an inherently difficult task, and so independence cannot be easily drawn with our familiar blobs.

Furthermore, this is where we see probability coming into its own. Before, we could safely draw an analogy between probability and area in the plane. Every probability idea had a one-to-one analog with areas of blobs. As soon as we introduce concepts like independence, that analogy starts to break down. Conditional probability and independence are the ideas that make probability *something new*. We aren't just doing real analysis anymore.

Definition 2.8. Two events $A, B \subseteq S$ are **conditionally independent** given a third event C if $P(A \cap B | C) = P(A | C)P(B | C)$.

That is *the definition* of conditional independence, but we can give it a satisfying interpretation by noticing the following consequence:

$$\begin{aligned} P(A | B \cap C) &= \frac{P(A \cap B \cap C)}{P(B \cap C)} && \text{def of conditional probability} \\ &= \frac{P(A \cap B | C)P(C)}{P(B \cap C)} && \text{MC decomp} \\ &= \frac{P(A | C)P(B | C)P(C)}{P(B \cap C)} && \text{def of conditional independence} \\ &= \frac{P(A | C)\cancel{P(B \cap C)}}{\cancel{P(B \cap C)}} && \text{MC decomp} \\ &= P(A | C). \end{aligned}$$

If you took C out of the picture, it may or may not be the case that A and B are independent. But once you know C , bringing in B as well teaches you nothing new about A . Given C , A and B don't care about one another.

Example 2.16. Conditioning on C can *remove* dependence between events. Suppose C is the event that a patient truly has COVID. Consider two diagnostic tests administered to the same patient:

$$A = \{\text{rapid test is positive}\}, \quad B = \{\text{PCR test is positive}\}.$$

The outcome of the tests depends on two factors: the true disease status and some random test-specific error. A and B are *dependent* because they are both influenced by the true disease status of the same patient. We can represent this with a *dependency graph*: $A \leftarrow C \rightarrow B$. The common dependence with C creates a *marginal* link between A and B . Absent knowledge of their actual health, if you knew the patient's result on one test, it should teach you *something* about what they are likely to get on the other test: $P(A | B) \neq P(A)$. However, if we already know that the patient is sick (C occurs and we know that), then *also* knowing one of the test results doesn't teach you anything new about the other test: $P(A | B \cap C) = P(A | C)$. Knowing B when you already know C is overkill. So A and B are not *marginally independent*, but they become *conditionally independent* given C . Conditional on C the only remaining randomness is the idiosyncratic error of the tests, which are unrelated.

Example 2.17. Conditioning can also *create* dependence between two unconditionally independent events A and B . Suppose we flip two separate coins. Let $A = \{1\text{st flip} = H\}$, $B = \{2\text{nd flip} = H\}$, and $C = \{1\text{st flip} = 2\text{nd flip}\}$. The two coin flips do not interfere with one another, so A and B are independent:

$$P(A \cap B) = P(A)P(B) = 0.5 \times 0.5 = 0.25.$$

It turns out that A and C are also independent:

$$P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{P(HH)}{P(HH \cup TT)} = \frac{P(HH)}{P(HH) + P(TT)} = \frac{0.25}{0.25 + 0.25} = 0.5 = P(A).$$

By the same reasoning, B and C are also independent. However, conditional on C , A and B become very dependent:

$$P(A | B \cap C) = P(1\text{st flip} = H | 2\text{nd flip} = H, 1\text{st flip} = 2\text{nd flip}) = 1 \neq P(A | C).$$

So, as soon as you know the two flips are the same, A and B become linked in a way they weren't *unconditionally*.

Definition 2.9. Events $A_1, A_2, \dots, A_n \subseteq S$ are **jointly** (or **mutually**) **independent** if the joint probability of *every* sub-collection factors into the product of marginals. So for any $j = 1, 2, \dots, n$, we can take any sub-collection $A_{i_1}, \dots, A_{i_j}$ of size j and get

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_j}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_j}) = \prod_{k=1}^j P(A_{i_k}).$$

Clear as mud?

It's really not as bad as all that. Consider three events $A, B, C \subseteq S$. We call these three events *jointly* independent if the following four statements are all true:

$$\begin{aligned} P(A \cap B) &= P(A)P(B) \\ P(A \cap C) &= P(A)P(C) \\ P(B \cap C) &= P(B)P(C) \\ P(A \cap B \cap C) &= P(A)P(B)P(C). \end{aligned}$$

So if the joint probability of *every* sub-collection factors into the product of marginals, we call the events jointly dependent. If instead of three events we had four, then we'd have to check all the pairs of events, *and* all the trios of events, *and* then the big four-way intersection. You need 'em all. If only the *pairs* factor but nothing else, we call the set of events **pairwise independent**. You can be pairwise independent but not jointly independent. This is the case for A , B , C in Example 2.17 (check!).

2.7 Summary of probability fundamentals

We are studying probability spaces:

- **(sample space)** a nonempty set S containing all possible outcomes of a random phenomenon;
- **(events)** subsets $A \subseteq S$;
- **(probability measure)** a function $A \mapsto P(A)$ that assigns probabilities to events.

We require by fiat that all probability measures satisfy these axioms:

- **(total measure 1)** $P(S) = 1$;
- **(nonnegativity)** $P(A) \geq 0$ for any $A \subseteq S$;
- **(countable additivity)** if $A_1, A_2, A_3, \dots \subseteq S$ is a sequence of pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

We can deduce from these axioms several downstream consequences:

- **(probability of the empty set)** $P(\emptyset) = 0$;
- **(complement rule)** $P(A) = 1 - P(A^c)$;
- **(monotonicity)** if $A \subseteq B$, then $P(A) \leq P(B)$;
- **(inclusion/exclusion)** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$;
- **(total probability)** if a collection of sets $B_1, B_2, B_3, \dots$ partitions S , then for any $A \subseteq S$:

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i) = \sum_{i=1}^{\infty} P(A | B_i)P(B_i);$$

- **(counting)** if S is finite and all outcomes are equally likely, then $P(A) = \#(A)/\#(S)$.

In these respects probability behaves similarly to other *measures* like length, area, and volume in Euclidean space. But the analogy starts to break down when we introduce conditional probability and independence. These ideas are the *special sauce* that sets probability apart as its own distinct branch of mathematics:

- **(conditioning)** $P(A | B) = P(A \cap B)/P(B)$, assuming $P(B) > 0$;
- **(marginal/conditional decomposition)** $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$;
- **(Bayes' theorem)** $P(A | B) = P(B | A)P(A)/P(B)$;
- **(independent events)** $P(A \cap B) = P(A)P(B)$, implying that $P(A | B) = P(A)$.

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